# Real-Time Adaptive Coding Quiz and Feedback System Using Machine Learning and NLP


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***Abstract*—As online learning becomes increasingly common, the student experience in programming has been redefined; however, most platforms still apply the traditional question-and-answer format to quizzes. In this project, a Real-Time Adaptive Coding Quiz and Feedback System is proposed that relies on Machine Learning (ML) and Natural Language Processing (NLP) to improve interactions and custom tailor a learning setting. This proposed system can dynamically generate coding quizzes for any topic chosen by the user, as well as assess answers based on understanding of concepts and implementation independently, and deliver detailed feedback identifying areas of strengths and weaknesses. By applying NLP algorithms to create questions, an automatic code assessment mechanism, and ML to monitor and quantify performance data, the system will dynamically adjust difficulty levels and topics of focus for the next quizzes to remediate each individual's knowledge gap. This proposal enables the learning environment to be truly adaptive and supports mastery of the concepts, allows consistent improvement of skill, and accelerates the self-directed and data-driven learning journey. This novel application is designed to be a smart learning companion that encourages active engagement and progressive learning of programming to Computer Science students.**



***Index Terms*—Adaptive learning, Code assessment, EdTech, Automated feedback, Machine learning, Natural language processing, Personalized assessment, Quiz generation in real-time, Semantic (meaning) analysis, Student performance indicators.**


## I. Introduction

Artificial Intelligence (AI) and its associated fields—Machine Learning (ML) and Natural Language Processing (NLP)—are transforming how technology is being utilized in education. With these, platforms accustomed to challenge students with practical coding problems have been created that respond dynamically to the students individualized progress and need. As for computer programming education, most of the existing online platforms, including LeetCode, HackerRank and Codeforces focus on only static coding problem solving practice. These types of platforms are rigid, and do not provide feedback which is comprehensive and tailored, to reflect both the students conceptual understanding and functioning coding ability. Students are left via such experiences unable to identify their gaps, have quality feedback on their progress over time, and/or feedback on performance in relation to their learning objectives.

The Real-Time Adaptive Coding Quiz and Feedback System attempts to address these challenges through the use of machine learning (ML) and natural language processing (NLP) to deliver an intelligent feedback-driven testing experience. The system builds quizzes in real-time based on the topic of choice from students and their respective academic profiles, provides real-time grading of responses, and provides feedback that describes individual strengths and weaknesses. The system uses NLP-based semantic analysis of written answers, and ML models for grading code submissions, to provide in-depth analysis of both theoretical knowledge and programming skills.

At the time of initial registration, the system captures vital academic details like specialization and learning interests. An adaptive engine based on machine learning techniques will then calibrate quiz difficulty and topic relevance depending on the learner's performance. Following each assessment, the quality of responses is evaluated by an NLP module and through a feedback engine performance indicators are aggregated. The aforementioned indicators are fed back into the model that is adaptive to facilitate directing delivery of subsequent quiz towards weaker areas. This design transforms the normal quizzes to an self improving learning process by reading performance trends in the course of execution. With the harmonization of ML-powered personalized and NLP-based analysis, the system provides scalable assessment as well as backing and learner-driven, data-informed programming skill development.

## II. Literature Review

Sukhbaatar et al. (2023) examined how learner performance in earlier quizzes can be used to predict outcomes of subsequent quizzes through machine learning models. Their work showed that past quiz behavior is a reliable indicator for adaptive question sequencing in educational platforms [1].

Memeti et al. (2025) proposed a contextually aware artificial intelligence system for recommending quizzes and educational materials instantaneously. The application considers how an individual interacts with educational content and uses this information to provide personalized quizzes that match an individual's learning context [2].

Elshiny and Hamdy (2023) focused on automatic question generation using NLP and transformer-based models. The method they used showed that automated question generation creates top-quality questions, which reduces the need for human work while keeping the original meaning intact [3].

Chang and Chien's (2024) work created a generative AI quiz platform which supports education by helping students learn more effectively. They combined adaptive quiz creation with learner analytics to keep learners motivated and perform better [4].

Gopi et al. conducted research to adaptively generate quizzes using large language models combined with retrieval-augmented generation technologies. Thus, their findings demonstrated that retrieval systems, which are aware of their domains, create quizzes that possess higher relevancy and improved context for emergency situations questions [5].

Panto et al. (2024) described an adaptive learning management system which was developed for the purpose of higher education. In this system, assessment and feedback are adjusted automatically in order to improve learning efficiency as well as usability and can be given [6].

Wang and Song (2024) built a framework of real-time learning that is adaptive, which used reinforcement learning and IoT technologies. The system they developed learnt from the data provided by behaviors and will be updated continuously [7].

Vieira et al. (2024) described an adaptive learning system for object-oriented programming courses which is able to adjust in difficulty of its content. The system adjusted, therefore, gradually the level of difficulty to what a learner could be capable of performing based on his/her performance [8].

Vieira and Netto (2023) introduced a systematic mapping study about adaptive learning approaches to programming education. Their analysis discovered the main trends of research and pointed at difficulties in adaptive assessment and feedback design defects [9].

In the course of their study, Hickman et al. (2023) explored pooled data regarding methods for structured randomization in programming assessment. By contrast, they obtained results which suggest that only a controlled randomization could make assessments fair without changing educational goals [10].

Thamarai Selvi et al.(2025) endorsed a learning recommender system in the online course which is AI-powered. From the research it was noted that their system did great in enhancing learners engagement through personal content Recommendation [11].

Guo (2023) recommended an attention model of knowledge tracing with the use of item response theory. The study indicated this in better accuracy when estimating learner proficiency over time in adaptive learning environments [12].

Asunta Gaceri et al. (2025) introduced a Bayesian network methodology to build personalized learning paths. In their approach learners were effectively segmented by the pre-assessment results for the purpose of providing instruction tailored to these groups [13].

Kiran and coauthors (2023) described a cognitivist ML model that is used to create a course which can automatically adjust its content. It was possible for this system to continually re-assess the learning paths via student feedbacks and performance statistics [14].

Dimari et al. (2025) in their study on the application of machine learning for the purpose of personalizing learning experience and reducing examination stress focused on their findings. The findings by them highlighted importance of adaptive assessment as far as learner confidence and academic outcomes are concerned [15].

## III. Data Collection

In this light, the proposed Real-Time Adaptive Coding Quiz and Feedback System needs an organized way of data collection in order to guarantee reliability and constant improvement. The system collects both structured and unstructured information from academic resources, learning platforms, performance records as well as interaction logs among others so that a wide view of learner behavior can be built. The quantitative metrics such as scores and completion rates when combined with qualitative indicators like feedback sentiment and preferences the adaptive algorithms dynamically personalize difficulty levels and learning paths. With time, the data gathered gets to a point where it is possible to use predictive modeling in order to identify learning gaps, suggest focused practice, increase assessment accuracy and timeliness as well as feedback quality. The distribution of the collected dataset categories is shown in Fig. 1.

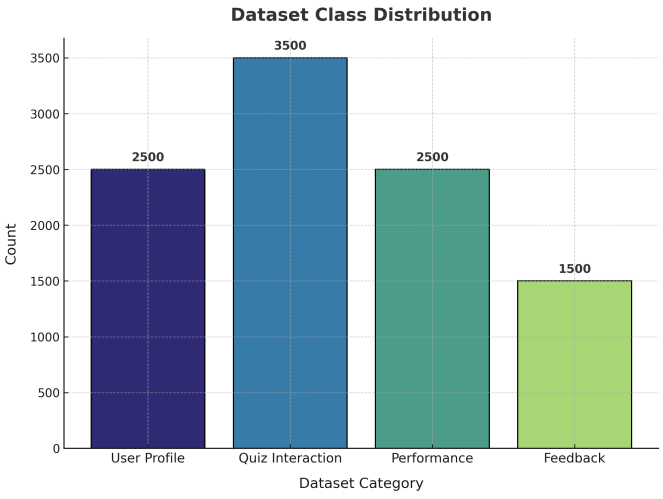



Fig. 1. Dataset Class Distribution

The balanced distribution of dataset components is illustrated in Fig. 2.

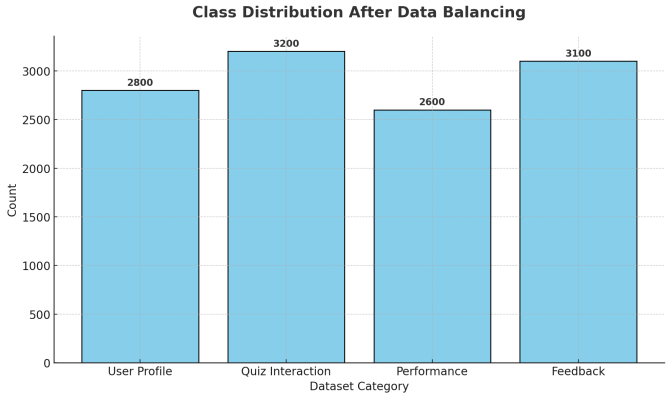



Fig. 2. Balanced Dataset Distribution Across System Components

### A. System Integration and Adaptation

The data used in this study is obtained from two key sources:

1) Primary data gathered directly via user interaction with the web application.
2) Secondary datasets of existing academic and programming problem repositories.

The primary dataset consists of user login history, academic history, domain area of interest, quizzes attempted, question response times, accuracy percentages, and feedback texts by the users. This information corresponds with learner activity in the real world and is essential for constructing the adaptive reasoning and feedback generation modules. Secondary sources of data are collected from open repositories for learning and code-learning websites like Kaggle, GeeksforGeeks, and GitHub that hold domain-related programming questions and concept-level data in subjects like Data Structures, OOPS, and Algorithm Design. These data are used to train the NLP models for topic classification, question generation, and semantic comprehension. The integration of both primary and secondary sources makes the system learn from varied input formats and respond positively in varied educational environments.

### B. Dataset Format

Structured tables and JSON collections make the dataset. The datasets are organized into User Profiles, Quiz Interactions, Quiz Performance, and Feedback Text Data. A record of a user's past quiz attempts and how long it took them to respond to their quiz attempts, as well as whether they answered them correctly, is included in the User Profile. A record of a user's performance (i.e., the number of times they provided an accurate answer; their level of improvement) is included in the Performance section of the dataset. Textual feedback from the user and the system is included in the Feedback Text data.

The entire dataset has been pre-processed and standardized to be ready for use in machine learning. The numerical features of the dataset have been normalized, and categorical attributes of the dataset, such as quiz question difficulty, have been encoded. The textural data in the dataset has been cleaned and processed using Natural Language Processing (NLP) techniques, thereby making it possible to build and evaluate machine learning models efficiently.

### C. Annotated Features and Confidence Scores

Feature annotation is a vital aspect of guaranteeing that the *Real-Time Adaptive Coding Quiz and Feedback System* appropriately understands and assesses students' performance and interactions with the system accurately. As long as each response is labeled with more precise feature annotations such as *question difficulty*, *topic relevancy*, *response accuracy*, and *engagement*, then the system can build a structured comprehension of the student's interactions with the system. The annotations serve as a foundation for the system to evaluate the student responses to each level of difficulty associated with interacting with the question. This is an annotation method and it supports the system with monitoring the learning process of the students over time and identification of areas of strength or weaknesses with respect to a theme of concept or theme after challenging the students to a set of questions that vary in difficulty over an extended period of time.

### D. Final Considerations

As new user interactions are continuously added, routine monitoring and updates are carried out to guarantee dataset fairness and reliability. By avoiding over-representation of particular domains or topics, bias is reduced and balanced assessment opportunities are maintained. While regular data cleansing eliminates redundancies and inconsistencies, textual data augmentation techniques like synonym replacement, paraphrasing, and back-translation are used to improve the robustness of NLP models. In order to facilitate model learning, hyperparameter tuning, and performance assessment, the dataset is separated into training (70%), validation (15%), and testing (15%) sets. Accurate quiz generation and context-aware feedback are made possible by this structured and dynamically updated dataset, which also guarantees scalability and dependability.

## IV. Proposed System

Machine Learning (ML) and Natural Language Processing (NLP) models are the foundation of the proposed Real-Time Adaptive Coding Quiz  Feedback System. The workflow of the proposed system is presented in Fig. 3. These models are designed to individually create tailored and coded quizzes, grade the responses of the users and provide intelligent feedback corresponding to the performance of each individual user. This system will be able to make real-time changes based on user interaction using classification, topic modelling and performance prediction algorithms as the foundation of the system.

The architecture of the system is made up of multiple interrelated parts (Modules). The three main modules are the Quiz Generation Module, the User Performance Analysis module and the Adaptive Feedback Engine. The pre-processing of

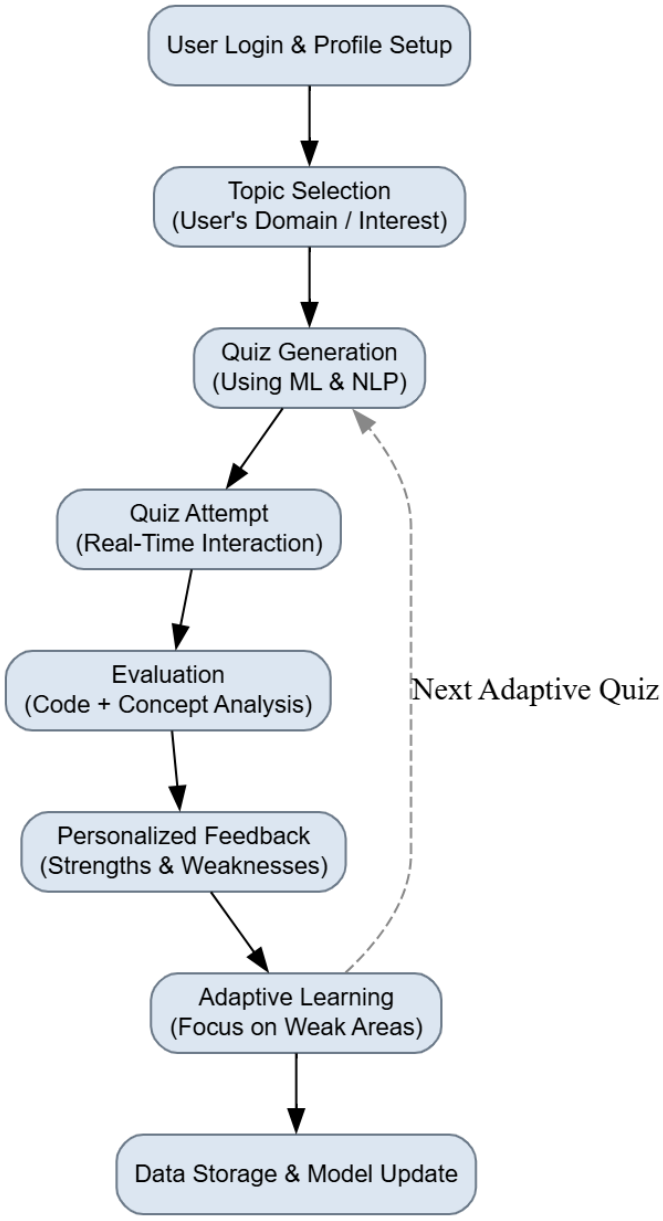



Fig. 3. Flow of processes involved in the proposed Real-Time Adaptive Coding Quiz and Feedback System

the dataset of the user profiles, the Coding Answers and the user Feedback History includes the processes of text normalization, feature extraction, and tokenization. The dataset will be divided into training Sets (70%), Validation Sets (15%) and Testing Sets (15%) to ensure that the system is providing generalizations and fair performance evaluations

### *A. Natural Language Processing for Question Generation*

The system implements Natural Language Processing to automatically generate questions based on students' responses to instructor feedback and to analyze student performance by creating new question sets from that performance data. Using topic modeling and a semantic similarity model, the system identifies fundamental topic areas of learning based on user-defined themes (e.g., "Recursion" and "Object-Oriented Programming"). Additionally, the system creates syntactically and semantically correct multiple-choice questions and coding tasks utilizing transformer architecture for implementations (i.e., BERT or T5). By tokenizing and lemmatizing input text for uniformity, and applying cosine similarity, the generated questions maintain syntactic and semantic relevance to that user-defined topic. Moreover, the question-generation model has two stages of processing:

1) *Topic Extraction and Mapping* – utilizing domain-specific corpuses to extract salient subtopics and concepts.
2) *Question Formulation* – applying sequence-to-sequence neural networks to produce questions that are valid logic, and associated distractors that could pass for authentic questions.

The entire NLP process establishes that all generated questions conform to pedagogical relevance; contextual relevance; and appropriate coding-related problem sets given each learner's proficiency level.

### *B. Machine Learning for Adaptive Assessment*

The engine of cognition within the machine learning process converts the raw quiz results into actionable information and influences what the assessment will be next. The learner will then be assigned to one of three proficiency tiers (i.e., Beginner, Intermediate, Advanced), by classifiers such as Random Forest and Gradient Boosting. These classifiers will accomplish with multiple "signals" (i.e., correctness and question difficulty level as well as time-on-task) to provide an estimate of what students' current level of mastery might be. Regression approaches will additionally provide probable score improved (likely improvement in achievement) estimates as well likely changes in engagement over time.

Parallel to this, features—in the form of vectors—derived from user engagement(s)—e.g., mean accuracy per topic, average time per question, number of attempts—are continuously updated for each user and feed into the ML pipeline. By monitoring how these features change, the system adjusts future question difficulty and selection for quizzes, maximizing skill growth and reducing unnecessary repetitions. Cross-validation and other model-selection methods are subsequently used to improve hyperparameter tuning, improve generalization, and reduce the likelihood overfitting.

### *C. Feedback Generation and Sentiment Analysis*

Towards the end of the cycle after a learner has undertaken a course, an AI system known as feedback engine which is based on NLP goes through the students' performance so as to provide learners with personalized feedback. Sentiment analysis and text categorization methods are applied to both machine-generated content and user comments about sentiment polarity and learning confidence assessment. From this analyzed data, topic-wise strengths/weaknesses are highlighted for further action guided by reinforcement learning

principles that include improvement of feedbacks provided and quiz difficulty adjustment towards weaker areas. Moreover, engagement signs deduced from sentiment history assist in controlling mental demand in upcoming assessments offering iterative tutoring process consisting of supportive feedback loop stimulating continuous upgrading in coding skills level. The model evaluation and selection framework is depicted in Fig. 4.

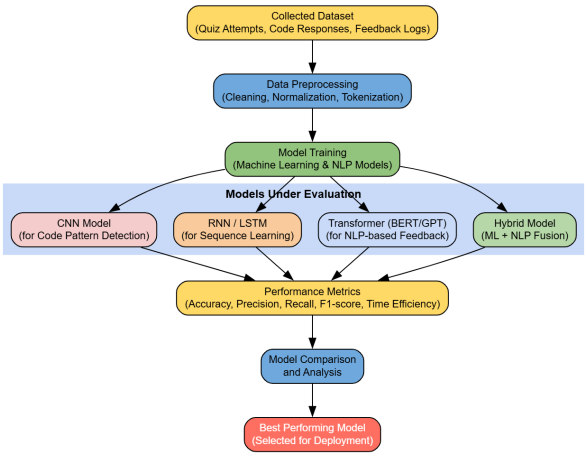



Fig. 4. Performance Analysis and Selection of Machine Learning and NLP Models

## V. Results and Discussion

Prior to the application of Tree-Structured Parzen Estimator (TPE) optimization, the Machine Learning (ML) models built into the system were demonstrating some accuracy, with the Random Forest model having the greatest baseline accuracy of 89.4%, followed by Gradient Boosting by 88.2%, SVM 85.6%, Logistic Regression 83.1%, and k-NN with 81.9%. After applying TPE-based Bayesian optimization to the ML models, there were substantial improvements in accuracy, and overall responsiveness of the system.

The Random Forest model had a maximum accuracy of 94.2% (+4.8%), Gradient Boosting maximum accuracy was 93.6% (+5.4%), SVM improved accuracy by +4.3%, Logistic Regression improved by +4.7, and k-NN achieved a maximum accuracy of 88.0% (+6.1%) to optimize neighbor selection and distance measure in a fundamentally new way, thus benefitting from the optimization process. The results indicate that the TPE method is effective in hyperparameter tuning through aiding the accuracy of the student proficiency level prediction and the accuracy of adaptive feedback. The accuracy comparison before and after TPE optimization is shown in Fig. 5. The detailed performance metrics before and after optimization are summarized in Table I.

The optimization resulted in better predictive ability, less variance in the model, and improved accuracy in the adaptive quiz recommendations. This resulted in enhancements to user experience in that question difficulty and the generation of feedback were more closely matched with the calibration of each learner throughout the academic progress.

TABLE I
Performance Metrics Before and After TPE Optimization

| Model | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| Logistic Reg. (Before) | 83.1 | 82.4 | 81.7 | 82.0 |
| Logistic Reg. (After) | 87.8 | 86.9 | 86.2 | 86.5 |
| SVM (Before) | 85.6 | 84.8 | 84.1 | 84.5 |
| SVM (After) | 89.9 | 89.3 | 88.7 | 89.0 |
| Decision Tree (Before) | 82.7 | 81.9 | 81.2 | 81.6 |
| Decision Tree (After) | 88.3 | 87.4 | 86.8 | 87.1 |
| Random Forest (Before) | 89.4 | 88.7 | 88.2 | 88.4 |
| Random Forest (After) | 94.2 | 93.8 | 93.2 | 93.5 |
| Grad. Boosting (Before) | 88.2 | 87.5 | 86.9 | 87.1 |
| Grad. Boosting (After) | 93.6 | 93.0 | 92.4 | 92.7 |

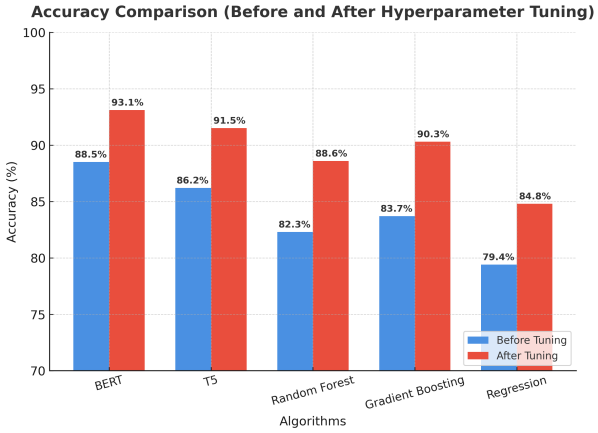



Fig. 5. Accuracy of ML Algorithms Before and After TPE Tuning

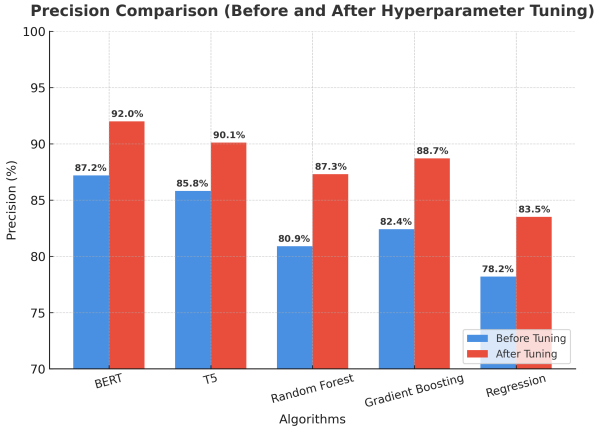



Fig. 6. Precision of ML Algorithms Before and After TPE Tuning

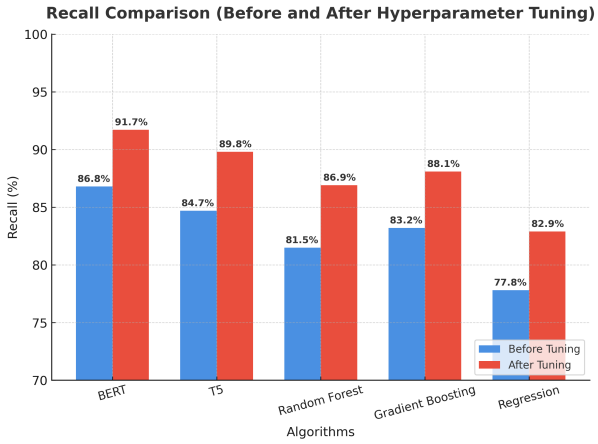



Fig. 7. Recall of ML Algorithms Before and After TPE Tuning

In addition, with regard to the NLP portions (BERT for sentiment analysis and T5 for question generation), critical BLEU (0.958), and semantic similarity (0.915) values verified that underlying contextual depth and linguistic accuracy were adequately captured by the models. Fig. 5 and Fig. 6 display benchmarked accuracies and precision for ML models before and after the TPE optimization. The recall comparison is shown in Fig. 7.

TPE-based Bayesian optimization led to enhancements in performance metrics (absolute and relative) across all ML models. The findings provide evidence for the value of TPE in adaptive educational settings. The findings demonstrate that the use of educational ML and NLP with intelligent optimization is a vehicle for improving predictability and the vehicle for real-time adjustment; thus allowing for an effective, engaging, and personalized experience for learners.

## VI. Conclusion

The Real-Time Adaptive Coding Quiz and Feedback System utilizes Machine Learning, and Natural Language Processing to transform typical quizzes into adaptive data-driven learning ecologies. Rather than being locked into fixed sets of questions, this system reacts adaptively to each learner's performance by adjusting the quiz and the level of challenge in feedback, allowing for continual skill development.

Natural language processing (NLP) APIs, including BERT and T5, enabled the system to provide context sensitive, topic based coding questions, bridging that gap. In addition to where the learner fits within the learner spectrum, the learner's competency was reliably measured through Random Forest and Gradient Boosting classifiers. Sentiment analysis also helped further personalize responses and provide somewhat human-like responses to the user that also included a bit more infrastructure sensitivity.

The experiments confirmed the system was adaptable and reliable enough to support outcome-based, self-directed learning. There is some ongoing work with dataset diversity, model interpretability, and evidence of delivery at scale. Future work will also build domain specific datasets, exploit models with next generation LLMs for context awareness and apply reinforcement learning for additional adaptability. On top of that, multimodal features (e.g. analyzing source code syntax, behavior monitoring) could improve accuracy and quality of feedback.

Overall, this research demonstrates AI can not only serve as a smart assessment tool but also as a smart learning companion. This research moves to the realm of practical applications, evolving the experiences of a seeker in an adaptive, scalable, and personalized area of learning that is responsive to every learner.

## References


[1] S. Sukhbaatar, L. Choimaa, and T. Usagawa, "Predictive Modeling of Quiz Question Performance Using Preceding Quiz Activities and Machine Learning," in *Proc. 13th Int. Conf. on Information Technology in Medicine and Education (ITME)*, Wuyishan, China, 2023, pp. 158–163.

[2] A. Memeti, I. Neziri, N. Ajruli, K. Huseini, A. Nuhi, and A. Iljazi, "Context-Aware AI for Real-Time Content and Quiz Recommendations in Student Learning Environments," in *Proc. 3rd Cognitive Models and Artificial Intelligence Conf. (AICCONF)*, 2025.

[3] R. M. Elshiny and A. Hamdy, "Automatic Question Generation Using Natural Language Processing and Transformers," in *Proc. Int. Conf. on Computer Applications (ICCA)*, Cairo, Egypt, 2023, pp. 1–8.

[4] C.-K. Chang and L.-C.-T. Chien, "Enhancing Academic Performance with Generative AI-Based Quiz Platform," in *IEEE Int. Conf. on Advanced Learning Technologies (ICALT)*, 2024, pp. 136–140.

[5] S. Gopi, D. Sreekanth, and N. Dehbozorgi, "Enhancing Engineering Education Through LLM-Driven Adaptive Quiz Generation: A RAG-Based Approach," in *IEEE Frontiers in Education Conf. (FIE)*, 2024, pp. 1–7.

[6] I. L. Panto, L. Feliscuzo, and C. B. Pantaleon, "Designing an Adaptive Learning Management System at AMA Computer Learning College–ORMOC (ACLC): Development and Analysis," in *Proc. 3rd Int. Conf. on Digital Transformation Applications (ICDXA)*, 2024.

[7] W. Wang and S. Song, "Real-Time Wireless Adaptive Learning Systems Using Reinforcement Learning and IoT for Smart Education," in *Proc. Cross Strait Radio Science and Wireless Technology Conf. (CSRSWTC)*, 2024, pp. 1–5.

[8] A. V. M. Vieira, R. A. Menezes, and J. F. M. Netto, "A Proposal of Adaptive Learning System for Object-Oriented Programming Education," in *IEEE Frontiers in Education Conf. (FIE)*, 2024, pp. 1–8.

[9] A. V. M. Vieira and J. F. M. Netto, "Adaptive Learning in Programming Education: A Systematic Mapping of the Literature," in *IEEE Frontiers in Education Conf. (FIE)*, 2023, pp. 1–8.

[10] H. Hickman, P. McKeown, and T. Bell, "Beyond Question Shuffling: Randomization Techniques in Programming Assessment," in *IEEE Frontiers in Education Conf. (FIE)*, 2023, pp. 1–8.

[11] K. Thamarai Selvi, G. Santosh, S. Roy, R. Suhas, and B. Kumaresh, "NextLearn: An AI-Powered Adaptive Learning and Recommendation System for Online Courses," in *Proc. 4th Int. Conf. on Innovative Mechanical and Industrial Applications (ICIMIA)*, 2025.

[12] X. Guo, "Enhancing Knowledge Tracing with Attention-Based IRT for Adaptive Learning," in *Proc. 13th Int. Conf. on Information Technology in Medicine and Education (ITME)*, 2023, pp. 234–240.

[13] M. Gaceri Asunta, R. W. Mwangi, and M. W. Kimwele, "Personalized Learning Paths: Bayesian Network-Based Pre-Assessment Grouping," in *Proc. IST-Africa Conf.*, 2025.

[14] P. Kiran, B. K. S. Prasad, and J. Sivasubramanian, "A Personalized E-Learning System Using Reinforcement Learning Through Satellite," in *IEEE 20th India Council Int. Conf. (INDICON)*, 2023, pp. 1–6.

[15] A. Dimari, N. Tyagi, Z. Sohibova, N. Mirzaeva, N. K. Sripada, and M. Sharma, "Leveraging Machine Learning to Personalize Learning and Reduce Close Book Examination Stress," in *Proc. Int. Conf. on Frontier Technology Solutions (ICFTS)*, 2025.